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# Install OpenCV
!pip install opencv-python-headless -q

import cv2
import numpy as np
import matplotlib.pyplot as plt
from skimage import data

# Load a sample image
image = data.astronaut()

# Convert RGB to BGR for OpenCV
image_bgr = cv2.cvtColor(image, cv2.COLOR_RGB2BGR)

# Get image dimensions
rows, cols = image_bgr.shape[:2]

# Translation values
tx = 150    # move right by 150 pixels
ty = 80     # move down by 80 pixels

# Translation matrix
M = np.float32([[1, 0, tx],
                [0, 1, ty]])

# Apply translation
translated = cv2.warpAffine(image_bgr, M, (cols, rows))

# Convert back to RGB for display
original_rgb = cv2.cvtColor(image_bgr, cv2.COLOR_BGR2RGB)
translated_rgb = cv2.cvtColor(translated, cv2.COLOR_BGR2RGB)

# Display images
plt.figure(figsize=(12,5))

plt.subplot(1,2,1)
plt.imshow(original_rgb)
plt.title("Original Image")
plt.axis("off")

plt.subplot(1,2,2)
plt.imshow(translated_rgb)
plt.title("Moved (Translated) Image")
plt.axis("off")

plt.show()
```

Original Image



Moved (Translated) Image

